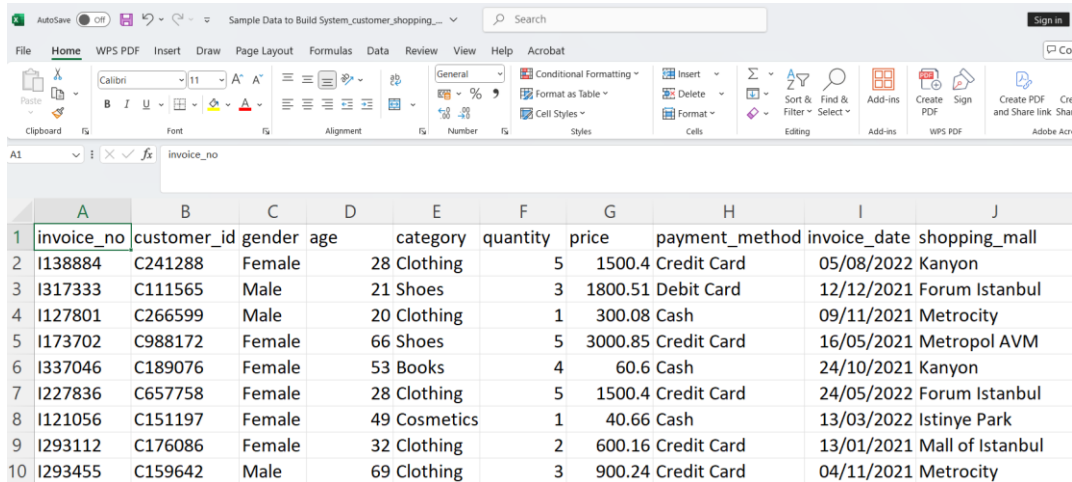


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CSC 112 – BNO1

MACHINE PROBLEM – CUSTOMER SHOPPING DATABASE

Solution:



	A	B	C	D	E	F	G	H	I	J
1	invoice_no	customer_id	gender	age	category	quantity	price	payment_method	invoice_date	shopping_mall
2	I138884	C241288	Female	28	Clothing	5	1500.4	Credit Card	05/08/2022	Kanyon
3	I317333	C111565	Male	21	Shoes	3	1800.51	Debit Card	12/12/2021	Forum Istanbul
4	I127801	C266599	Male	20	Clothing	1	300.08	Cash	09/11/2021	Metrocity
5	I173702	C988172	Female	66	Shoes	5	3000.85	Credit Card	16/05/2021	Metropol AVM
6	I337046	C189076	Female	53	Books	4	60.6	Cash	24/10/2021	Kanyon
7	I227836	C657758	Female	28	Clothing	5	1500.4	Credit Card	24/05/2022	Forum Istanbul
8	I121056	C151197	Female	49	Cosmetics	1	40.66	Cash	13/03/2022	Istinye Park
9	I293112	C176086	Female	32	Clothing	2	600.16	Credit Card	13/01/2021	Mall of Istanbul
10	I293455	C159642	Male	69	Clothing	3	900.24	Credit Card	04/11/2021	Metrocity

Step 1: 1NF (Normal Form)

- ✓ All values are atomic.
- ✓ The table is already in 1NF.

Step 2: 2NF

- ✓ The primary key is **invoice_id**, so there's no partial dependency. Hence, the table is already in 2NF.

Step 3: 3NF

- ✓ It is in the Second Normal Form (2NF).
- ✗ There is a transitive dependency like:
 - **invoice_no** → **customer_id** → **gender**
 - **invoice_no** → **customer_id** → **age**
 - **invoice_no** → **category** → **price**

Normalized Database

customers		
customer_id	gender	age
C241288	Male	21
C111565	Female	49
C266599	Male	69
C988172	Female	32
C189076	Female	53
C657758	Female	28
C151197	Male	20
C176086	Female	28
C159642	Female	66

invoices				
invoice_no	customer_id	invoice_date	shopping_mall	payment_method
I138884	C241288	8/5/2022	Kanyon	Credit Card
I317333	C111565	12/12/2021	Forum Istanbul	Debit Card
I127801	C266599	11/9/2021	Metrocity	Cash
I173702	C988172	5/16/2021	Metropol AVM	Credit Card
I337046	C189076	10/24/2021	Kanyon	Cash
I227836	C657758	5/24/2022	Forum Istanbul	Credit Card
I121056	C151197	3/13/2020	Istinye Park	Cash
I293112	C176086	1/13/2021	Mall of Istanbul	Credit Card
I293455	C159642	11/14/2021	Metrocity	Credit Card

products		
product_id	category	price
1	Clothing	1500.4
2	Shoes	1800.51
3	Clothing	300.08
4	Shoes	3000.85
5	Books	60.6
6	Clothing	1500.4
7	Cosmetics	40.66
8	Clothing	600.16
9	Clothing	900.24

invoice_items		
invoice_no	product_id	quantity
I138884	Male	21
I317333	Female	49
I127801	Male	69
I173702	Female	32
I337046	Female	53
I227836	Female	28
I121056	Male	20
I293112	Female	28
I293455	Female	66

1. Daily Sales by Mall

```
SELECT
i.invoice_date as date,
i.shopping_mall as mall,
SUM(ii.quantity * p.price) as total_revenue
FROM invoices as i
JOIN invoice_items as ii on i.invoice_no = ii.invoice_no
JOIN products as p on ii.product_id = p.product_id
JOIN products as p on ii.product_id = p.product_id
GROUP BY i.invoice_date, i.shopping_mall
ORDER BY i.invoice_date, i.shopping_mall;
```

date	mall	total_revenue
2020-03-13	Istinye Park	40.66
2021-01-13	Mall of Istanbul	1200.32
2021-05-16	Metropol AVM	15004.25
2021-10-24	Kanyon	242.40
2021-11-09	Metrocity	300.08
2021-11-14	Metrocity	2700.72
2021-12-12	Forum Istanbul	5401.53
2022-05-24	Forum Istanbul	7502.00
2022-08-05	Kanyon	7502.00

9 rows in set (0.02 sec)

2. Customer Spending by Age Group

```
SELECT
CASE
    WHEN c.age < 20 THEN '<20'
    WHEN c.age BETWEEN 20 AND 29 THEN '20s'
    WHEN c.age BETWEEN 30 AND 39 THEN '30s'
    WHEN c.age BETWEEN 40 AND 49 THEN '40s'
    WHEN c.age BETWEEN 50 AND 59 THEN '50s'
    ELSE '60+'
END AS age_group,
COUNT(DISTINCT c.customer_id) AS customers,
SUM(ii.quantity * p.price) AS total_spend,
ROUND(SUM(ii.quantity * p.price) / COUNT(DISTINCT
c.customer_id), 2) AS avg_spend_per_customer

FROM customers AS c
JOIN invoices AS i on c.customer_id = i.customer_id
JOIN invoice_items ii on i.invoice_no = ii.invoice_no
JOIN products p on ii.product_id = p.product_id
GROUP BY age_group
ORDER BY age_group;
```

age_group	customers	total_spend	avg_spend_per_customer
20s	4	20705.61	5176.40
30s	1	1200.32	1200.32
40s	1	40.66	40.66
50s	1	242.40	242.40
60+	2	17704.97	8852.49

5 rows in set (0.02 sec)